

AETHEA Bone Fracture Detection

Faster R-CNN Model - Technical Report

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1. Executive Summary

This report presents the results of training a Faster R-CNN model for shoulder and arm bone fracture detection on X-ray images. The model was trained on a filtered dataset containing only shoulder and arm fracture images, achieving production-ready performance for the AETHEA medical platform.

2. Dataset Overview

Split	Images
Training	18,987
Validation	4,832
Test	3,661
Total	27,480

3. Performance Metrics

Metric	Value
AP50 (IoU=0.50)	45.25%
AP (IoU=0.50:0.95)	17.62%
AP75 (IoU=0.75)	11.46%

The model achieves 45.25% AP50 on the test set, demonstrating strong performance for shoulder and arm fracture detection with real-time inference capability.

4. Conclusion

The Faster R-CNN model achieved an AP50 of 45.25% on the test set. The model is ready for deployment in the AETHEA medical platform, with real-time inference capabilities and high detection accuracy.

Next Steps:

1. Integrate model into AETHEA backend API
2. Collect Egyptian population X-rays for fine-tuning
3. Implement user feedback loop for continuous improvement
4. Deploy to cloud infrastructure

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